

GradeSense System Architecture

Vector views of frontend, backend, data, ML, WebSocket, and Render deployment

Curated, implementation-linked documentation. Code examples are selectable text, never screenshots.

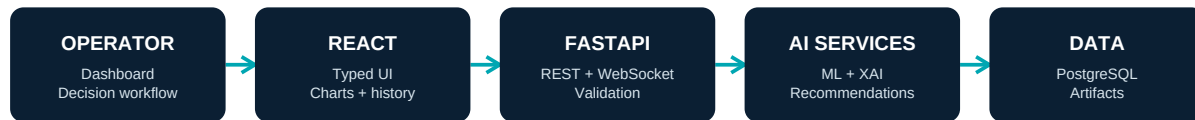
Candidate

Name: Bhavuk Mahajan

Candidate ID: 20484629

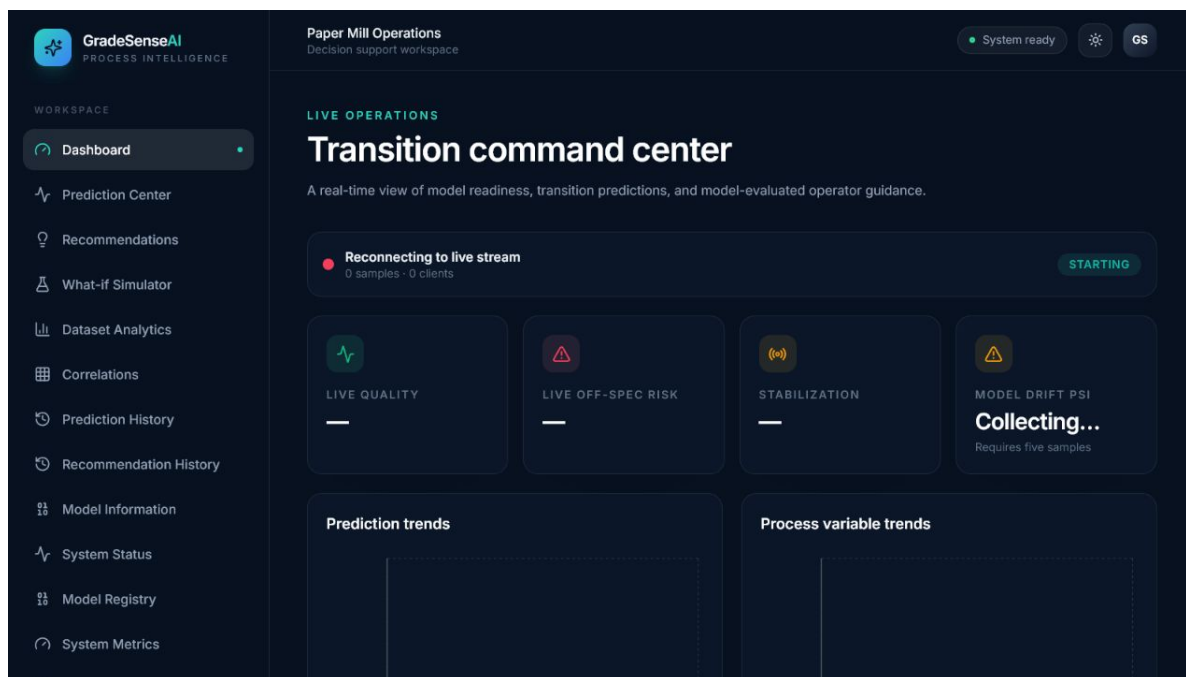


1. Overall Architecture

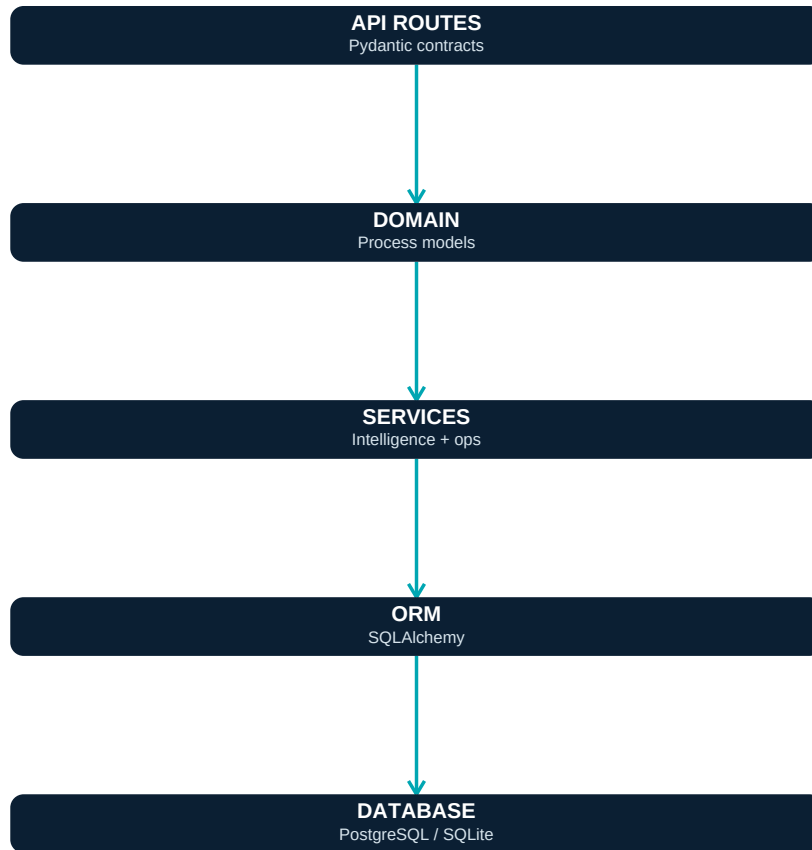


Quality	Architectural response
Traceability	Prediction, recommendation, decision, outcome and audit persistence
Safety	Constraint checks and human-in-the-loop advisory output
Governance	Immutable artifacts and registry promotion validation
Real time	Bounded stream worker and WebSocket broadcaster

2. Operator Workspace

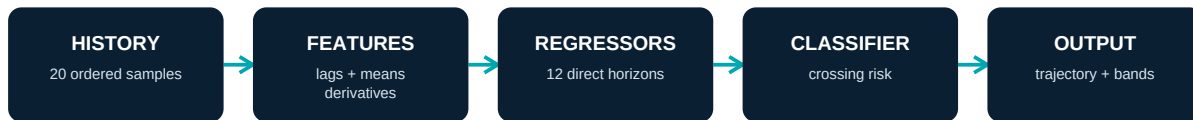


3. Backend Service Architecture



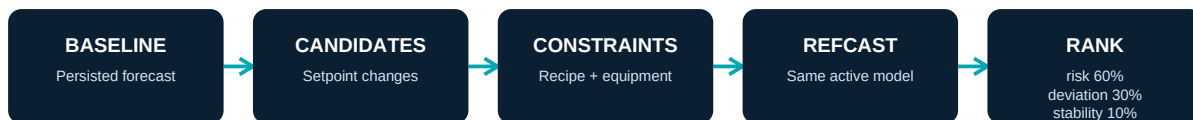
FastAPI lifespan validates model readiness, bootstraps the registry, records startup, and starts streaming. Shutdown stops the worker, records the event, and disposes database resources.

4. Machine-Learning Architecture



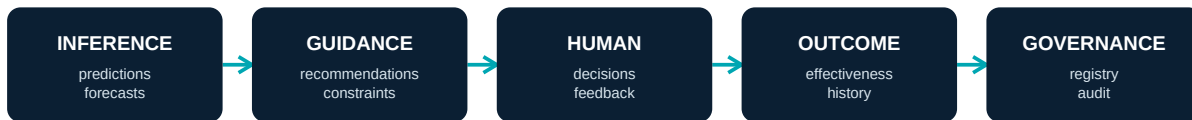
Family	Model	Output
Snapshot	Multi-output Random Forest	Quality, off-spec probability, stabilization
Sequential	HistGradientBoostingRegressor x12	Future Basis Weight trajectory
Crossing	GradientBoostingClassifier	Any >2.5% crossing in horizon
Uncertainty	Held-out residual envelope	Empirical 90% bands

5. Recommendation and Explainability Architecture



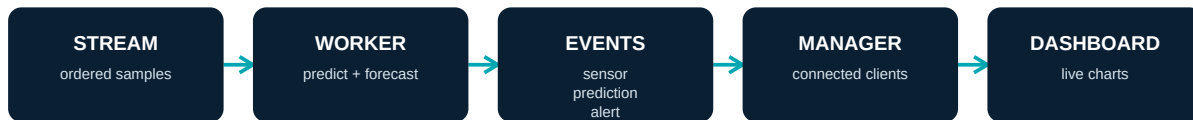
Local explainability perturbs one feature to its learned baseline and measures the prediction delta. Recommendation evidence combines forecast causes, historical trend, correlations, recipe-rule attribution and similar-transition outcomes. Sensitivity and correlation are explicitly non-causal.

6. Database and Audit Architecture



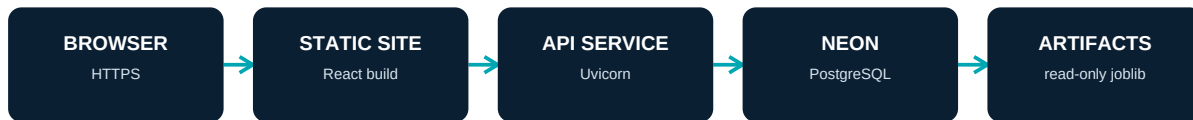
Relational entities retain searchable identifiers and timestamps. Nested trajectories, explanations, metrics and constraint evidence are stored as structured JSON where the data is naturally hierarchical.

7. WebSocket Architecture



Property	Design
Backpressure	Bounded queues and recent-sample deque
Lifecycle	Started/stopped by FastAPI lifespan
Channels	/ws/live and /ws/dashboard
Scale limit	In-process manager targets one API replica
Scale path	Shared broker and centralized metrics

8. Render Deployment



Render builds the frontend as a static site and the backend as a Python web service. Alembic migrations run during backend build; /health drives readiness. CORS and API origins are supplied through environment variables.